Discrete Math 2026 Final Solutions

1. **T(rue) or F(alse).** You don't have to give a reason, but it might get you half a point if you are wrong.

   The diameter of a graph is twice its radius F. This is true for very few graphs, such as paths. For cycles, say, the diameter and the radius are the same.

   ii. Graphs with the same degree sequence are isomorphic. F

   iii. The sequence $(3, 3, 3, 4, 5, 5, 5, 6)$ is graphic. T

   iv. Eulerian graphs have degree sequence consisting of even integers. T

   v. A graph with n vertices and $n-1$ edges is connected. F. Only if it is also acyclic.

   vi. Every tree has at least two leaves. F: The tree on one vertex doesn't. (But I thought it was T too. I think I probably said this in class. So this problem wasn't graded.)

   vii. All trees on n vertices are isomorphic. F. We show how to decide if two trees on n vertices are isomorphic.

   viii. A graph is connected if and only if it has a spanning tree. T

   ix. $K_{2,3}$ is planar. T

   x. There are non-planar graph that can be drawn on the sphere without crossings. F

2. Draw all non-isomorphic graphs with degree sequence $(3, 3, 3, 3, 3, 3, 6)$.

   Solution

   There are two, take either C_6 or two copies of C_3 and add a vertex adjacent to everything.

3. When can a directed graph be drawn with a (not necessarily closed) single line? Each directed edge must be drawn exactly once from head to tail.

   Solution

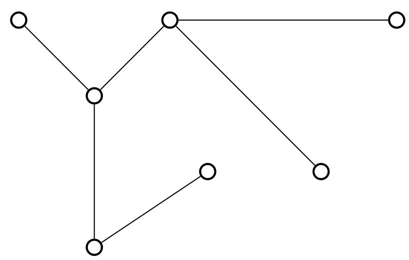
   The graph must be connected, and every vertex v except at most two must have $\deg^-(v) = \deg^+(v)$. If two do not satisfy this, then one, s, must have $\deg^-(s) = \deg^+(s) + 1$ and the other, t, must have $\deg^-(t) = \deg^+(t) - 1$.

4. How many arcs does the DeBruijn graph $D^3_{0,1,2}$ have? How many are loops?

Solution

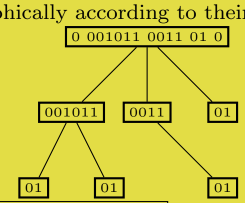
The graph has 9 vertices: $\{0,1,2\}^2$. Every vertex (a,b) has an arc to (b,c) for any c, so every vertex has out-degree 3. So there are $\boxed{\text{27 arcs}}$. The pairs (i,i) have loops, so among the arcs there are $\boxed{\text{3 loops}}$.

5. Recall that in determining it two trees are isomorphic we defined a code for each tree. Give the code for the following tree.



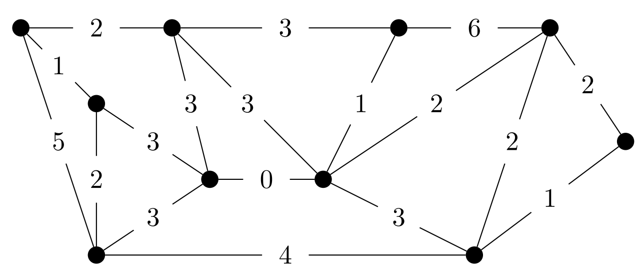
Solution

Finding the center in the middle of the path of length 4, it has three children. Ordering them lexicographically according to their codes,



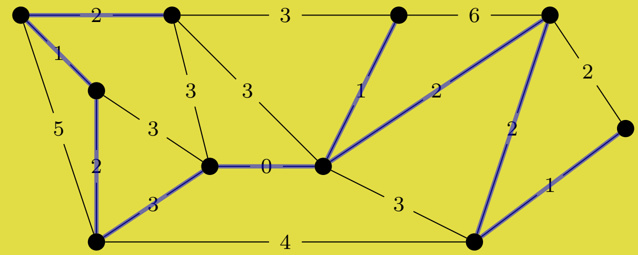
The tree's code is $\boxed{0\ \ 001011\ \ 0011\ \ 01\ \ 0}$ (spaces removed).

6. Find the weight of a minimum weight spanning tree for the graph shown.



Solution

$\boxed{14}$ Here is a minimum weight spanning tree.



7. (a) State Euler's formula.
   (b) Use it to show that K_5 is not planar.

Solution

i) For any connected plane graph with n vertices, m edges and f faces

$$n - m + f = 2.$$

ii) Assume there is a drawing of K_5 in the plane. By Euler's formula, the drawing has $f = 2 - n + m = 2 - 5 + 10 = 7$ faces. But each face has at least 3 edges, and each edge is in at most two faces, so the number of edges is at least $7 * 3/2 = 10.5$, which is not true. Thus K_5 can not be planar.

Proofs

Do **one** of the following problems. (Use the back of page if you need room.)

8. Recall that a tournament is an oriented graph that we get by orienting the edges of a complete graph. Show that any tournament has a vertex z such that for every other vertex u there is a directed path from u to z of length at most 2.

Solution

Let z be a vertex of maximum indegree. Let u be any other vertex that does not have an edge to z. As z has indegree at least that of u, and z is counted among the in-neighbours of u there is some vertex v that is an in-neighbour of z but not one of u. So $u \to v \to z$ is a directed path of length 2 from u to z.

9. Show that the maximum number of edges in a triangle-free graph on n vertices is $\lfloor n^2/4 \rfloor$.

Solution

We show this by induction on n. It is clearly true when $n \in 1, 2$. Let G be a triangle free graph on $n \geq 3$ vertices with the maximum number of edges. Let $e = uv$ be an edge of G, and G' be the graph we get from G by removing the vertices u and v. Any vertex $x \in V(G')$ can have at most one edge to u or v. So G can have at most $n-1$ edges more than G', which by the induction hypothesis has at most $\lfloor n - 2/4 \rfloor^2$ edges. So, when n is even, we have

$$|E(G)| \leq (n-2)^2/4 + n - 1 = \frac{n^2 - 4n + 4 + 4n - 4}{4} = n^2/4$$

and when n is odd, the floor removes one from each fraction giving the result.

10. Let V be a set of points in $\mathbb{R}^2$. Make a weighted graph (G, w) on V by putting, for every pair of points u, v in V, an edge uv in G with weight $w(u, v) = |u - v|$. Here $|u - v|$ is the (Euclidean) distance between the points in $\mathbb{R}^2$. Show that no minimum weight spanning tree of G has a vertex v with degree $\deg(v) \geq 7$.

Solution

Assume, towards contradiction, that a vertex v has $d \geq 7$ neighbours $u_1, \ldots, u_d$ in a spanning tree T of G. We may assume that v is drawn at the origin, and the u_i are ordered by the angle their arc to v makes with the positive x axis. There is some i such that the angle $u_i v u_{i+1}$ is at most $360^\circ/d < 60^\circ$. But then the angle at v is not the largest angle in the triangle $u_i v u_{i+1}$. So the distance $|u_i - u_{i+1}|$ is less than at least one of the distances $|v - u_i|$ and $|v - u_{i+1}|$. Thus there is a smaller weight spanning tree of G that uses $u_i u_{i+1}$ instead of one of these edges. This is a contradiction.